I. Use Case Description	
Use Case Name	Reasoning About Action Feasibility from Physical Properties
Use Case Identifier	UC-ActionFeasibility-01
Source	LACAI Lab, Department of Computer Science, Rensselaer Polytechnic Institute
Point of Contact	Hyejin Park, Department of Computer Science, Rensselaer Polytechnic Institute, parkh12@rpi.edu
Creation / Revision Date	September 5, 2026
Associated Documents	Nothing at this stage

II. Use Case Summary	
Goal	Develop an ontology that captures relationships among objects, their physical properties, and action requirements so that a system can reason about whether an object is suitable for a given action and explain why.
Requirements	The ontology should represent objects, actions, physical properties, and action requirements, as well as the relationships among them. It should capture both enabling and blocking conditions for an action and support reasoning about whether an object satisfies the physical requirements of a given action. The representation should also support an explanation of why an action is feasible or infeasible based on the relevant physical properties.
Scope	The initial scope is limited to everyday object manipulation in kitchen environments, where objects are used to perform common actions such as cutting, opening, containing, stirring, or peeling. The ontology will focus on physical properties of objects, relevant object parts, and action requirements that determine whether an action is feasible or infeasible. Broader environments and more complex activities are outside the initial scope but may be incorporated in later extensions.
Priority	High — representing action feasibility from physical properties is central to the intended reasoning task and provides the foundation for later extensions of the ontology.
Stakeholders	Primary stakeholders are researchers in knowledge representation, robotics, and embodied AI who study object affordances and action feasibility. Secondary stakeholders include developers of intelligent systems that need to reason about object use from physical properties.
Description	This use case considers a system that reasons about whether an everyday object can be used to perform a particular manipulation action in a kitchen environment. Given an object, an intended action, and relevant physical information, the ontology connects the object's physical properties and parts with the requirements of the action. The system can then determine whether the action is feasible or infeasible and provide an explanation based on the properties that enable or block the action.
Actors / Interfaces	**Primary Actor – User/Researcher:** Submits queries about whether a given object can support a particular action and receives the resulting judgment and explanation. **System – Ontology-Based Reasoning System:** Matches the physical properties of the object against the requirements of the action and infers whether the action is feasible or infeasible. **Secondary Actor / Interface – Knowledge Base/Data Sources:** Supplies structured information about objects, their parts and physical properties, and action requirements.
Pre-conditions	
Post-conditions	
Triggers	
Performance Requirements	
Assumptions	
Open Issues	

III. Usage Scenarios

Provide at least two usage scenarios that flesh out the requirements outlined in the summary, including identification of requirements specific to any envisioned ontology or semantically-driven service or application. Scenarios should be described as narrative, with supporting diagrams as appropriate. In an Agile process, every user story relevant to the use case should be included and elaborated/rolled up into one or more usage scenarios, with a clear mapping from the user story to the scenario it is integrated in or mapped to.

IV. Basic Flow of Events

Narrative: *Often referred to as the primary scenario or course of events, the basic flow defines the process/data/work flow that would be followed if the use case were to follow its main plot from start to end. Error states or alternate states that might occur as a matter of course in fulfilling the use case should be included under Alternate Flow of Events, below. The basic flow should provide any reviewer a quick overview of how an implementation is intended to work. A summary paragraph should be included that provides such an overview (which can include lists, conversational analysis that captures stakeholder interview information, etc.), followed by more detail expressed via the table structure.*

In cases where the user scenarios are sufficiently different from one another, it may be helpful to describe the flow for each scenario independently, and then merge them together in a composite flow.

Basic / Normal Flow of Events			
Step	**Actor (Person)**	**Actor (System)**	**Description**

V. Alternate Flow of Events

Narrative: *The alternate flow defines the process/data/work flow that would be followed if the use case enters an error or alternate state from the basic flow defined, above. A summary paragraph should be included that provides an overview of each alternate flow, followed by more detail expressed via the table structure.*

Alternate Flow of Events			
Step	**Actor (Person)**	**Actor (System)**	**Description**

VI. Use Case and Activity Diagram(s)

Provide the primary use case diagram, including actors, and a high-level activity diagram to show the flow of primary events that include/surround the use case. Subordinate diagrams that map the flow for each usage scenario should be included as appropriate

VII. Competency Questions

CQ1 — Feasibility judgment
Can this object be used to perform this action, and why?

- Example Answer: A metal spoon can be used to stir a liquid because it has sufficient rigidity and a graspable elongated part that can transmit motion to the liquid.
- Ontology Approach: The ontology represents classes such as *Object*, *ObjectPart*, *Action*, and *PhysicalProperty*, with relations such as *hasPart*, *hasProperty*, and *requiresProperty*. For a given object-action pair, the system retrieves the physical properties of the relevant object parts and compares them with the properties required by the action. For example, the rigidity, geometry, and graspability of a spoon can be matched against the requirements of *Stirring* to determine whether the action is feasible and explain why.

CQ2 — Alternative object discovery
Given an action, what unconventional objects could satisfy its physical requirements?

- Example Answer: A rigid plastic card may be used to cut a soft food item if its edge is sufficiently thin and stiff, even though the object is not designed as a cutting tool.
- Ontology Approach: The ontology starts from an *Action* such as *Cutting* and follows its *requiresProperty* relations to identify the required physical properties. It then retrieves *Object* or *ObjectPart* entities whose *hasProperty* values satisfy those requirements, regardless of their conventional function. In this way, a rigid plastic card can be inferred as a possible alternative tool when its thin, stiff edge satisfies the physical requirements for cutting a soft material.

VIII. Resources

Data Availability: The initial kitchen-domain action vocabulary is available from EPIC-KITCHENS-100. However, the physical-property and affordance relations required for this use case are not directly provided by the dataset and are currently being generated and verified through ongoing experiments. These derived annotations will serve as the primary knowledge source for constructing the ontology.

Knowledge Bases, Repositories, or other Data Sources

Data	Type	Characteristics	Description	Owner	Source	Access Policies & Usage
EPIC-KITCHENS-100	Remote dataset / repository	Egocentric kitchen videos with action annotations; verb and noun classes	Provides the initial kitchen-domain action vocabulary and examples of human-object interactions. The action annotations are used to identify common manipulation actions to be	EPIC-KITCHENS / University of Bristol and collaborators	EPIC-KITCHENS dataset website / annotation repository Link (full dataset): https://epic-kitchens.github.io/ Link (annotation	Publicly available for research use; Required for defining the initial kitchen action domain

			represented in the ontology.		dataset): https://github.com/epic-kitchens/epic-kitchens-100-annotations	
Derived Physical-Property and Affordance Annotations	Local / experimentally generated dataset	Structured annotations linking actions, objects, object parts, physical properties, action requirements, and feasible/infeasible outcomes	Provides the physical-property and affordance relations needed by the ontology. These annotations are generated and verified through ongoing experiments rather than provided directly by EPIC-KITCHENS.	Hyejin Park / LACAI Lab	Generated from the ongoing affordance-reasoning experiments using kitchen-domain actions derived from EPIC-KITCHENS	Currently internal and under development; Required for constructing the physical-property and action-requirement portions of the ontology

External Ontologies, Vocabularies, or other Model Services

Resource	Language	Description	Owner	Source	Describes/Uses	Access Policies & Usage
Word Net	English	Provides word senses, synonym sets, and lexical-semantic relations for English nouns and verbs. It can support normalization and sense distinction for action and object terms used in the ontology.	Princeton University	Princeton WordNet Link: https://wordnet.princeton.edu/	Action and object terminology derived from EPIC-KITCHENS and the experimental annotations	Freely available; Supporting / Optional

Other Resources, Service, or Triggers *(e.g., event notification services, application services, etc.)*

Resource	Type	Description	Owner	Source	Access Policies & Usage
(sensor or external service name)		*Include a description of the resource as well as availability, if applicable*	*Primary owner of the service*	***Application or service URL**; if subscription based, include subscription and any subscription owner*	

IX. References and Bibliography

List all reference documents – policy documents, regulations, standards, de-facto standards, glossaries, dictionaries and thesauri, taxonomies, and any other reference materials considered relevant to the use case

X. Notes

There is always some piece of information that is required that has no other place to go. This is the place for that information.

* LLM Use Statement:

I used ChatGPT (GPT-5.6 Sol) to help revise the English wording and organization of this document. All substantive content and ontology design decisions are my own.